RE07 Global Infrastructure Investment and Management

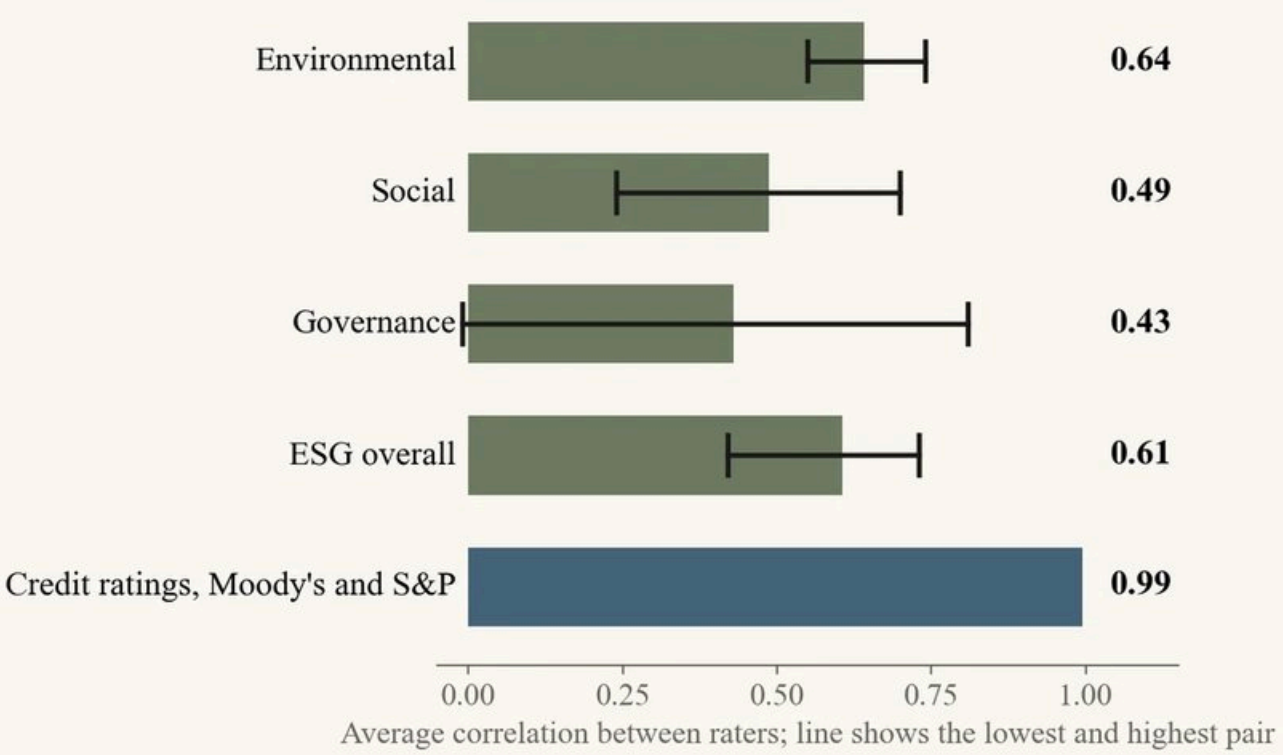
8. Sustainability, ESG and Emerging Trends

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Lecture Outline

Measuring sustainability
Climate and water risk in prices
Carbon and green labels in asset prices
Disclosure and supervision
Data centres, debris and the environment
Pricing sustainability and climate risk in long-lived assets
Case Study: Water Infrastructure and the Environmental Dimension of ESG (Andreas Jobst, Central Bank of the UAE)

ESG Ratings Disagree



Five raters: KLD, Sustainalytics, Vigeo-Eiris, RobecoSAM and Asset4, on a common sample of firms

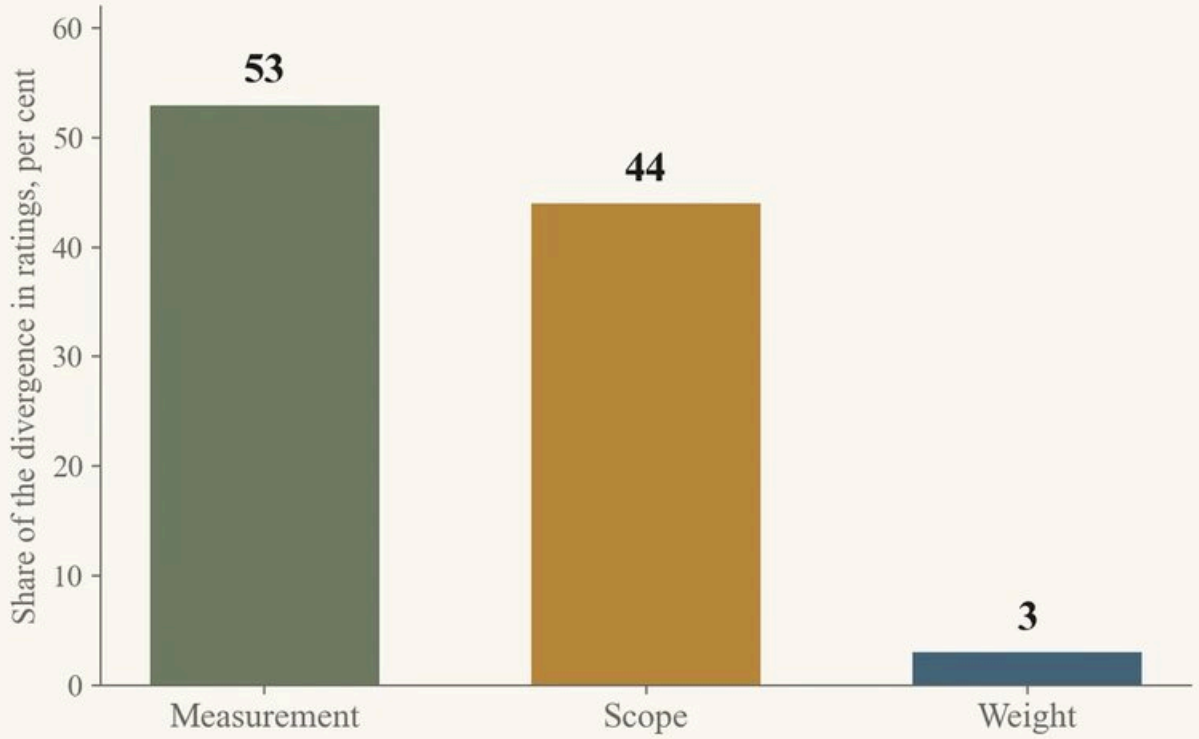
The correlation: 0.61 on average, and from 0.42 to 0.73 across the ten pairs of raters

Credit ratings: Moody's and Standard and Poor's correlate at 0.99

The dimensions: agreement is lowest on social and governance scores

Berg, Koelbel and Rigobon (2019), MIT Sloan Working Paper 5822-19, Table 2, page 7; credit-rating correlation from page 2. Averages are computed from the table.

Sources of Rating Divergence



Measurement: the same category is scored differently, 53 per cent of the divergence

Scope: raters cover different categories, 44 per cent

Weight: raters weight categories differently, 3 per cent

Water: all five raters score it, and the scores correlate at 0.33 on average

Berg, Koelbel and Rigobon (2019), MIT Sloan Working Paper 5822-19, pages 3 to 4 (decomposition) and page 18 (Table 5, water category). Regression-based decomposition.

Climate Risk in Long-Dated Asset Prices

Asset	Evidence	Size	What it shows
Coastal homes	Bernstein and co-authors (2019)	14.7 per cent discount at one foot of sea-level rise; 13.8 at two to three feet; 4.4 at six feet	Exposed homes sell below equivalent unexposed homes
Municipal bonds	Painter (2020); Goldsmith-Pinkham and co-authors (2019)	Present value of long-run cash flows 3 to 8 per cent lower	The yield effect sits in long-dated bonds and is absent at short maturities
Discount rates	Giglio, Kelly and Stroebel (2020)	Sign of the risk premium on climate hedges is disputed	Negative if the uncertainty is about climate, positive if it is about the economy

Giglio, Kelly and Stroebel (2020), NBER Working Paper 28226, pages 7 to 9, 16, 22 and 23. Bernstein and co-authors, Painter and Goldsmith-Pinkham and co-authors are reported through the survey.

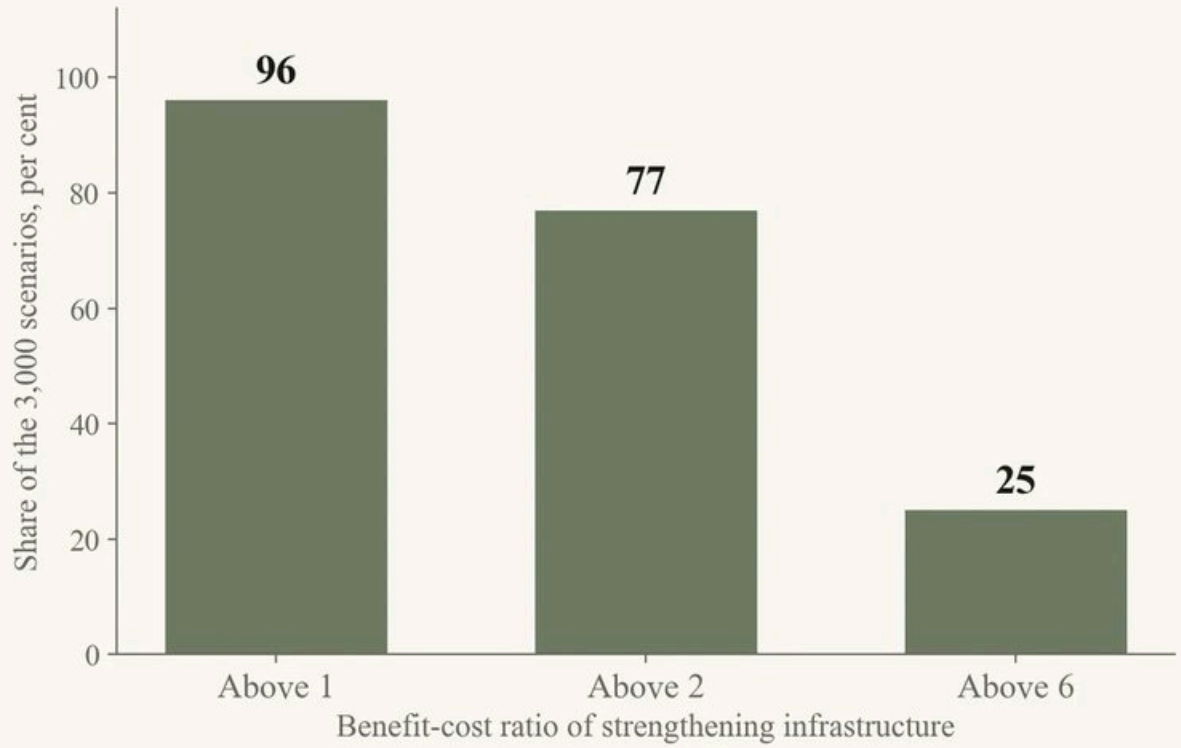
Water Risk Indicators

Baseline water stress, demand as a share of supply	Risk category	Score
Below 10 per cent	Low	0 to 1
10 to 20 per cent	Low-medium	1 to 2
20 to 40 per cent	Medium-high	2 to 3
40 to 80 per cent	High	3 to 4
Above 80 per cent	Extremely high	4 to 5

Thirteen indicators in three groups, covering quantity, quality and reputational risk, are combined into one overall score. Each is assigned a risk element, hazard, exposure or vulnerability, and risk is their product.

Kuzma and co-authors (2023), Aqueduct 4.0, WRI Technical Note, August 2023, pages 1 to 2, 10 and 11. Thresholds from Aqueduct 2.1 (Gassert and co-authors, 2014).

The Cost of Disrupted Infrastructure



The cost of disruption: at least USD 390 billion a year to households and firms in low and middle income countries

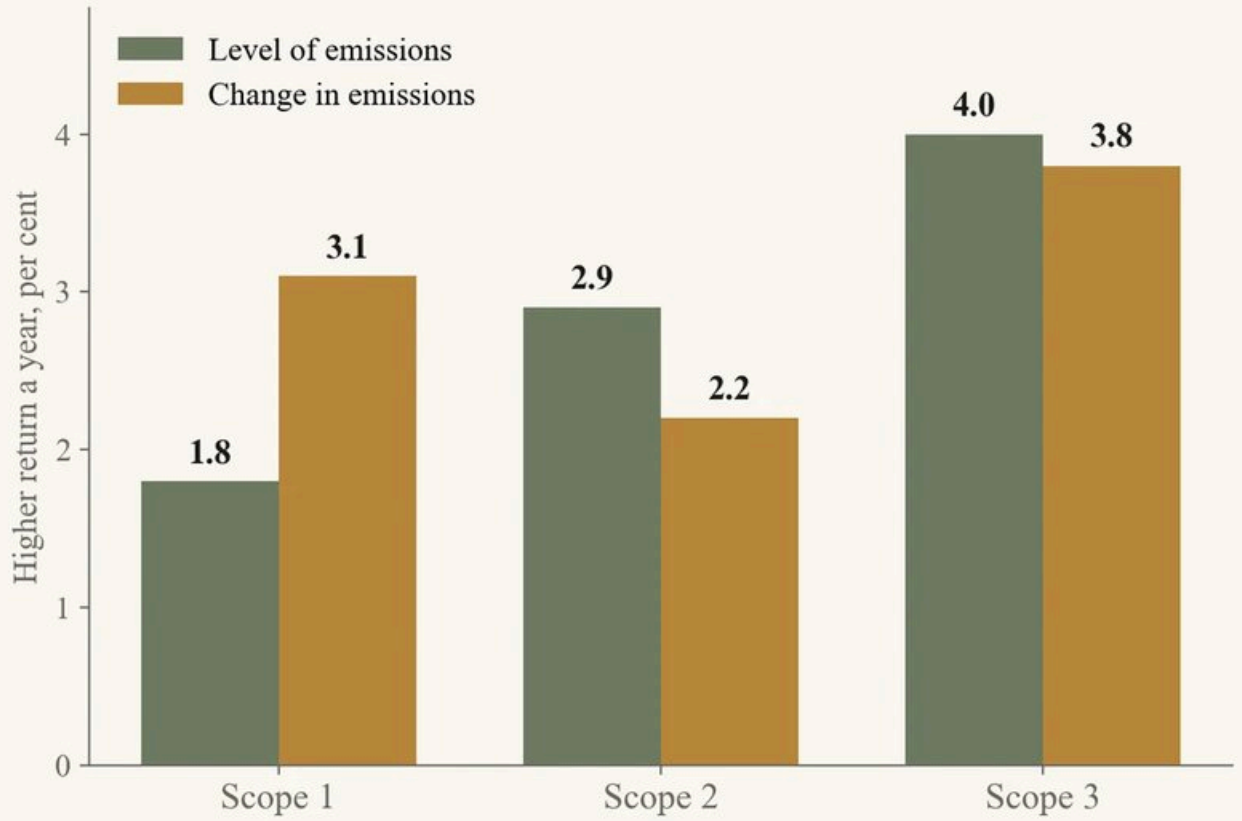
The cost of resilience: 3 per cent added to investment needs in new infrastructure

The return: USD 4 of benefit for each dollar, USD 4.2 trillion over the life of new assets

The spread: the ratio is above 6 in a quarter of 3,000 scenarios and above 1 in 96 per cent

Hallegatte, Rentschler and Rozenberg (2019), Lifelines: The Resilient Infrastructure Opportunity, World Bank, Overview, pages vii and 12. Pre-publication overview booklet.

The Carbon Premium in Stock Returns



The premium: firms with higher emissions earn higher returns, after the usual controls

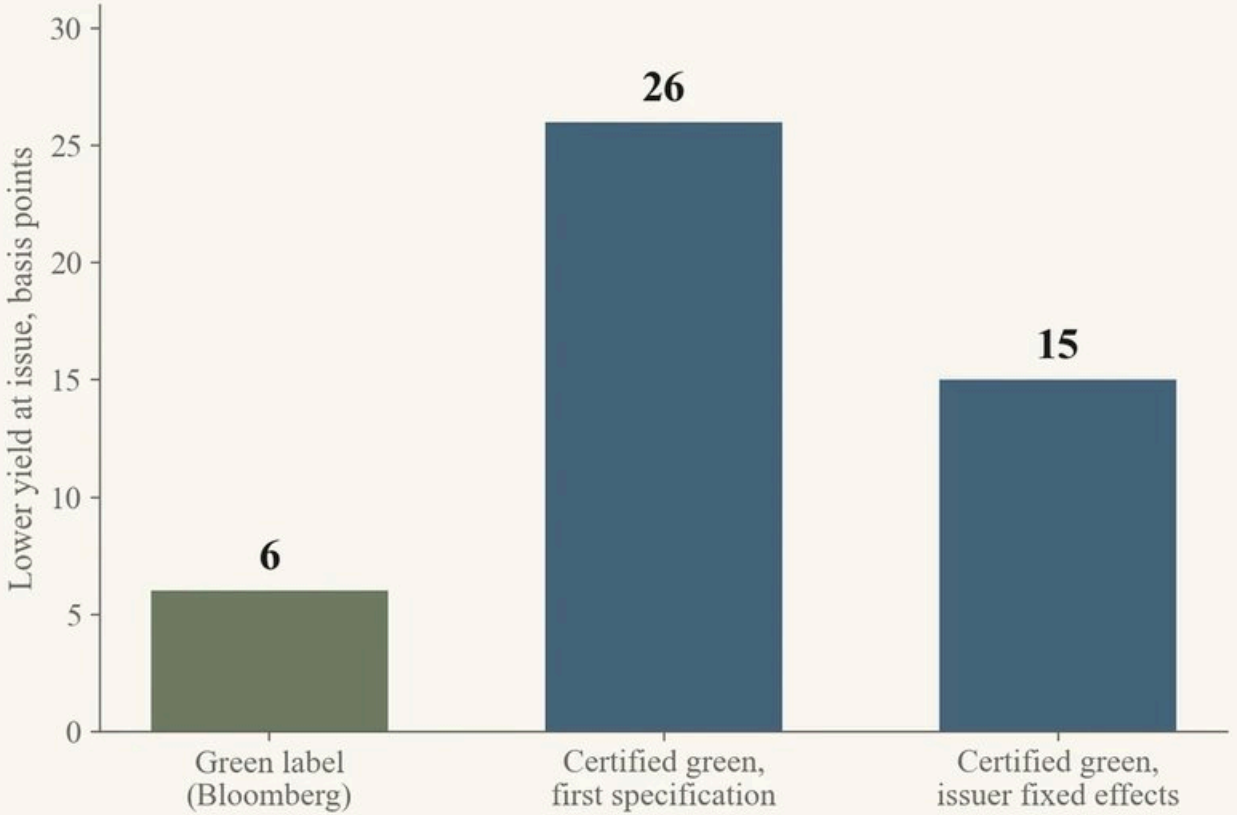
The size: one standard deviation of scope 1 adds 1.8 per cent a year, and of scope 3 adds 4.0 per cent

Intensity: emission intensity, the usual screening measure, has no significant effect

Timing: no premium appears in the 1990s, so awareness matters

Bolton and Kacperczyk (2020), NBER Working Paper 26968, pages 1 and 6 to 8, and Table 8 discussion, page 18. Annualised effect of a one-standard-deviation increase.

The Green Bond Premium



The framework: assets that give a nonpecuniary benefit are priced at a premium

Green municipal bonds: about 6 basis points lower yield, 0.6 percentage points of value at a ten-year duration

Certified green: 26 basis points lower in the first specification and 15 with issuer fixed effects

Ownership: green bonds are more closely held than ordinary bonds

Baker, Bergstresser, Serafeim and Wurgler (2018), NBER Working Paper 25194, pages 1 to 3, 17 and 18. After-tax yield at issue, US municipal bonds.

Climate-Related Disclosure: Four Pillars

Governance	Strategy	Risk management	Metrics and targets
Board oversight of risks and opportunities	Risks and opportunities over the short, medium and long term	Processes for identifying and assessing risks	Metrics used to assess risks and opportunities
Management's role in assessing and managing them	Impact on business, strategy and financial planning	Processes for managing risks	Scope 1, scope 2 and, if appropriate, scope 3 emissions
	Resilience of the strategy under scenarios, including 2 degrees Celsius or lower	Integration into overall risk management	Targets and performance against them

Transition risks are policy and legal, technology, market and reputation. Physical risks are acute, event-driven, or chronic, longer-term shifts. Water availability, sourcing and quality are named among the physical effects.

Task Force on Climate-related Financial Disclosures (2017), Final Report, June 2017, Figure 4, page 14, and pages 5 and 6. Wording shortened.

Water as a Financial Stability Risk

The meeting: 21 and 22 May 2026 at the Banco de España, convened by the OECD, the NGFS and CDP

The risk: macroeconomic and financial risks from water scarcity, variability and quality

The agenda: sectoral vulnerabilities, sovereign exposure, price stability, supervisory responses and insurance protection gaps

The aim: to advance understanding of water as a systemic financial stability risk

The programme: Scaling Water Resilience in Finance, run by UNEP FI and the OECD, plans a central bank joint statement drafted through the NGFS (152 members) and a showcase of market practice

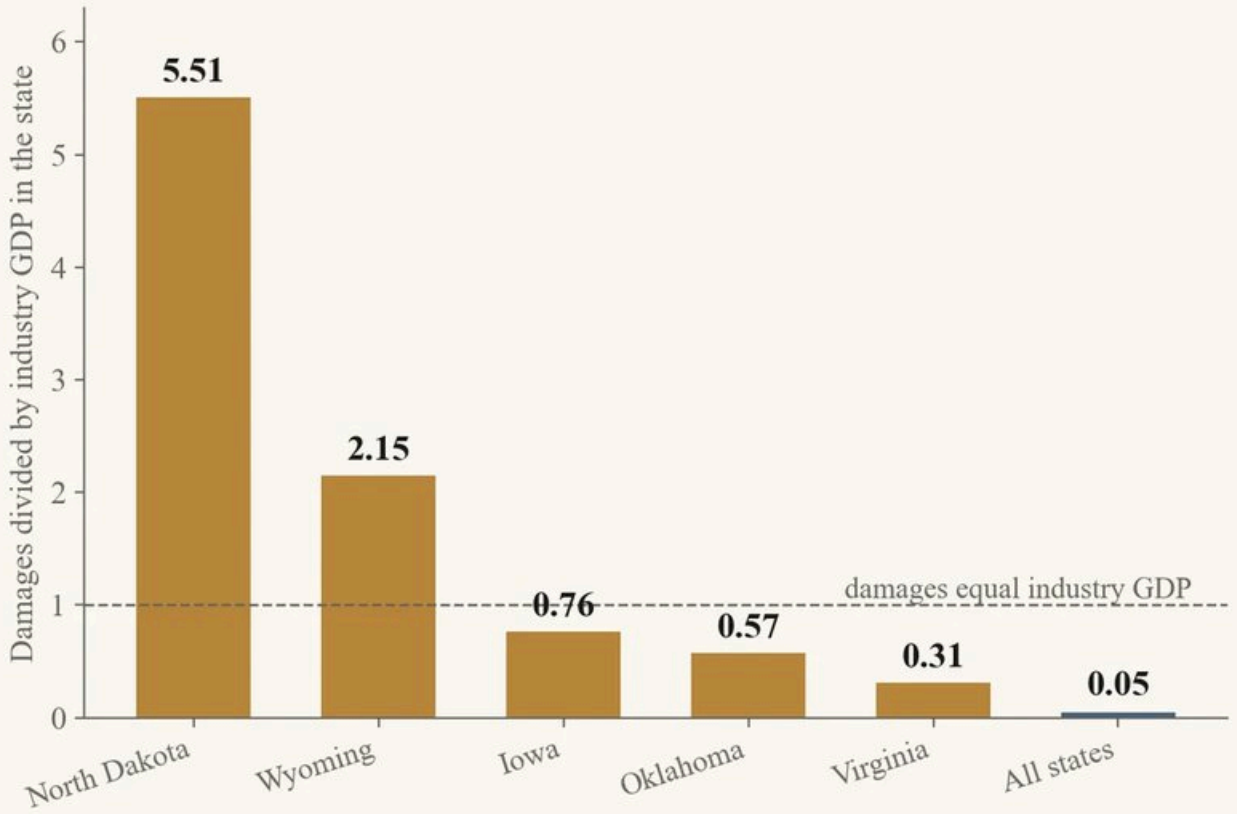
Banco de España (2026), press release of 20 May 2026, an announcement of the agenda and not a report of conclusions. Jobst, A. (2026), Scaling Water Resilience in Finance: Policy, Practice and Action, UNEP FI and OECD programme proposal, draft V.04.04.2026, pages 1 to 8, plans as at April 2026. NGFS membership page, 26 August 2026.

The Programme: Two Planned Outputs and a Timeline

	Joint statement on water risk in financial supervision	Water risk practice showcase
What it is	A statement drafted by central banks through the NGFS, extending climate and nature approaches to freshwater	A curated showcase of how banks, insurers, investors and rating agencies already handle water risk
What it is not	A commitment to any supervision, or a replacement for climate and nature frameworks	A compact, a requirement to change methods, or a supervisory expectation
Who endorses or contributes	Central banks and supervisors, NGFS members, regional leaders including the UAE Central Bank	Insurers, rating agencies, banks, investors and development finance institutions
Timing	Drafted in Q3 2026, published in Q4 2026	Curated in Q3 2026, launched in Q4 2026
After that	2027: OECD-led roundtables on supporting uptake by supervisors	2027: continued engagement through UNEP FI and OECD platforms

Jobst, A. (2026), Scaling Water Resilience in Finance: Policy, Practice and Action, UNEP FI and OECD programme proposal, draft V.04.04.2026, pages 4 to 6. Plans as at April 2026, not outcomes.

Data Centres and Environmental Damages



The estimate: 250 TWh in 2025, 5 to 6 per cent of US generation, and USD 25 billion of damages, within a range of USD 10 to 33 billion

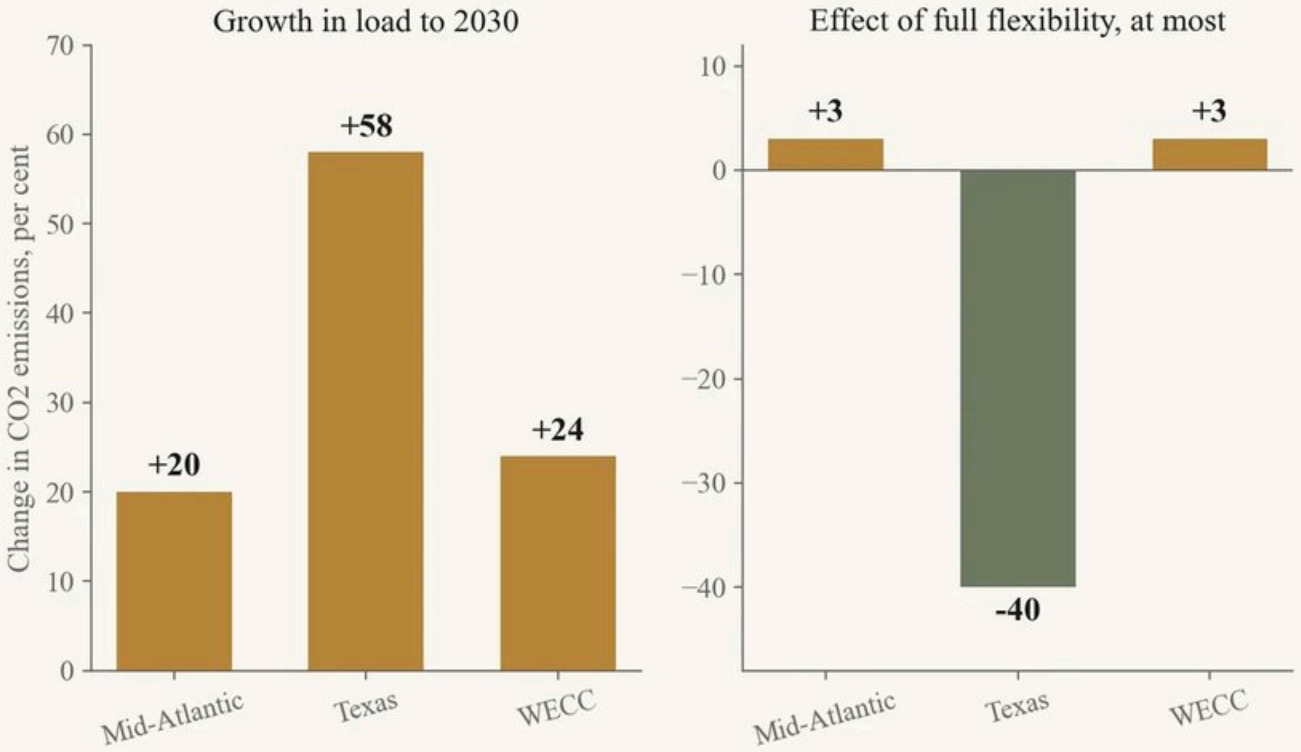
The incidence: Texas and Virginia bear 30 per cent of the total

The comparison: 5 per cent of industry GDP overall, and more than industry GDP in North Dakota and Wyoming

The valuation: carbon at USD 190 a tonne, or USD 50 in the alternative

Muller (2026), NBER Working Paper 35100, April 2026, abstract and Table 2. Gross external damages divided by state GDP of data processing, hosting and related services.

Flexible Data Centres and the Grid



The baseline: load growth to 2030 raises CO2 by 20, 58 and 24 per cent in the three regions

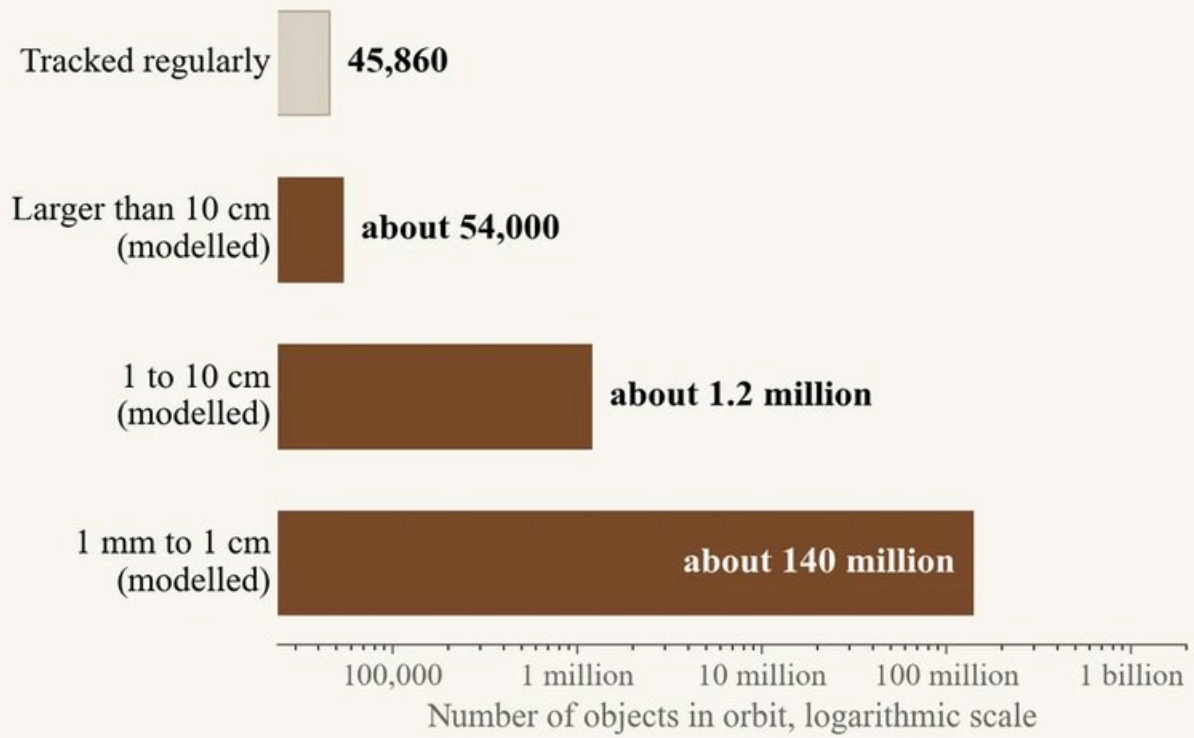
The cost: flexible load lowers system costs by up to 4, 5 and 2 per cent

The emissions: up to 40 per cent lower in Texas, up to 3 per cent higher elsewhere

The condition: the advantage disappears where renewables are dear

Knittel, Senga and Wang (2025), NBER Working Paper 34065, July 2025, pages 1 to 3, 8 and 12. Model of three US systems in 2030; 'at most' values from the scenarios.

Orbital Debris as an Environmental Externality



The stock: about 45,860 objects are tracked, and models estimate 140 million between 1 mm and 1 cm

The persistence: above 700 km decay takes centuries or longer; at lower altitudes decades, months or days

The dynamic: simulations show debris can become self-generating, with collisions creating more debris without new launches

The cost: Starlink satellites made over 300,000 collision-avoidance manoeuvres in 2025, from about 200,000 in 2024

OECD (2026), The Space Economy at a Glance 2026, pages 79 to 81, citing ESA (2026), Annual Space Environment Report, and operator reports. Counts by size are model estimates.

Three Models for Long-Lived Assets

Topic	Model	Prediction	Page
The price of sustainability	Pastor, Stambaugh and Taylor (2021)	Greener assets cost more and earn less; investors who care give up return	Pricing sustainability preferences
The value of a distant benefit	Green Book (2026), June 2026 review, gamma discounting	The schedule decides what a century of benefit is worth	Discounting long-lived assets
The payoff to protection	Expected annual loss with a protection standard	Net benefit peaks at a standard that moves with climate and discounting	The payoff to resilience

Each page opens on a generic setting. The case supplies the evidence to test them.

Pastor, Stambaugh and Taylor (2021); HM Treasury (2026); Freeman and Groom (2026); author's calculation. Interactive pages in web/.

Pastor, Stambaugh and Taylor (2021): Tastes and Prices

The setting: investors gain a nonpecuniary benefit from green holdings, in proportion to their taste

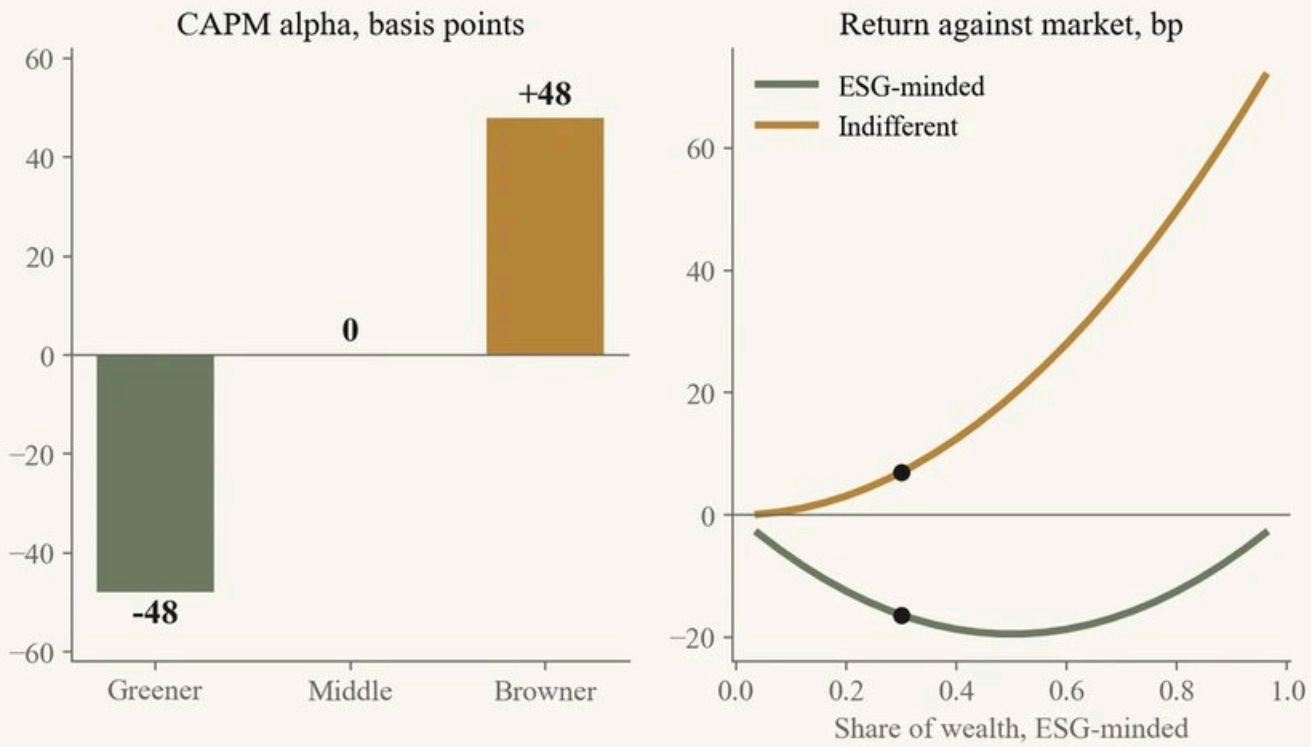
The price: expected return is the CAPM return less the average taste over risk aversion, times the ESG score

The portfolios: above-average investors tilt green, below-average investors tilt brown

The condition: if everyone has the same taste, all hold the market and the alphas remain

Pastor, Stambaugh and Taylor (2021), Journal of Financial Economics 142(2), 550–571; NBER Working Paper 26549, equations 4 to 10 and 14, pages 6 to 11.

The Price of a Preference



The alpha: minus 48 basis points a year on the greener asset and plus 48 on the browner one

The give-up: ESG-minded investors earn 16 basis points a year less than the market

The gain: indifferent investors earn 7 basis points more

The dispersion: the give-up is largest at equal wealth and zero when tastes are the same

Author's calculation from Pastor, Stambaugh and Taylor (2021), NBER Working Paper 26549, equations 9, 10 and 14; illustrative inputs (models_w7.py).

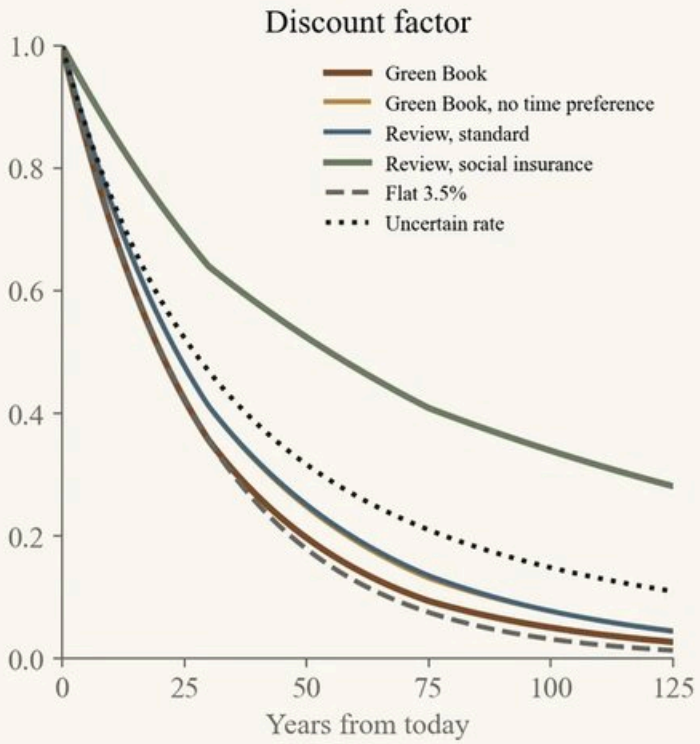
The Treasury Discount Rate and Its Schedules

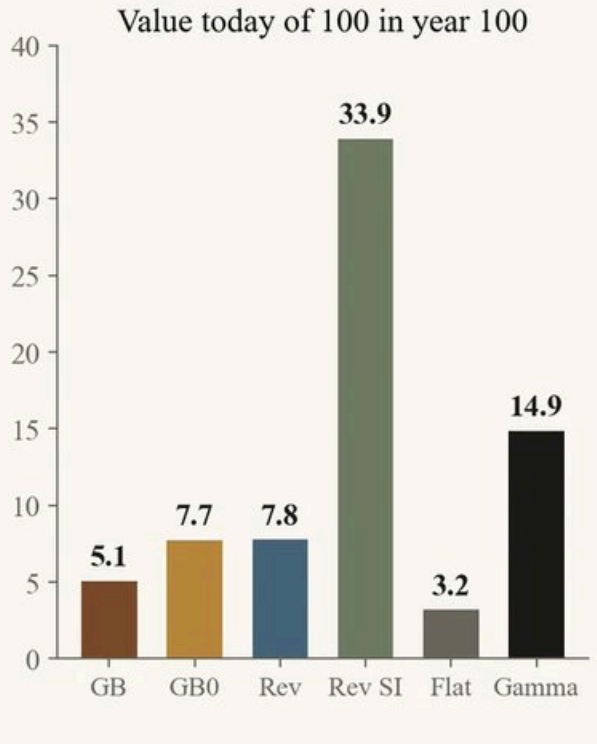
Forward rate, per cent	Years 1 to 30	Years 31 to 75	Years 76 to 125
Green Book, February 2026	3.5	3.0	2.5
Green Book, no pure time preference	3.00	2.57	2.14
Review, June 2026, standard projects	3.0	2.5	2.25
Review, June 2026, social insurance projects	1.5	1.0	0.75

An uncertain rate with a gamma distribution gives a forward rate that starts at the mean and falls with the horizon. The review is independent and is not Treasury policy.

HM Treasury (2026), Green Book supplementary guidance: discounting, Tables 3.A and 3.B, pages 12 and 13; Freeman and Groom (2026), Table 1, page 2 of 15.

The Value of a Distant Benefit





The range: 100 received in year 100 is worth 3.2 at a flat 3.5 per cent and 33.9 on the social insurance schedule

The Treasury: 5.1 on the Green Book schedule

The uncertain rate: 14.9 with a mean of 3 per cent and a standard deviation of 2

A level stream: a benefit-cost ratio from 1.5 to 2.4, a spread of 1.6, against 10.6 for the distant amount

Author's calculation from HM Treasury (2026), Tables 3.A and 3.B, and Freeman and Groom (2026), Table 1; gamma regime derived (models_w7.py). Illustrative project.

Expected Annual Loss and Protection

The curve: loss from an event with a return period of T years is the 100-year loss times (T/100) to the power s

Expected loss: the area under loss against annual probability

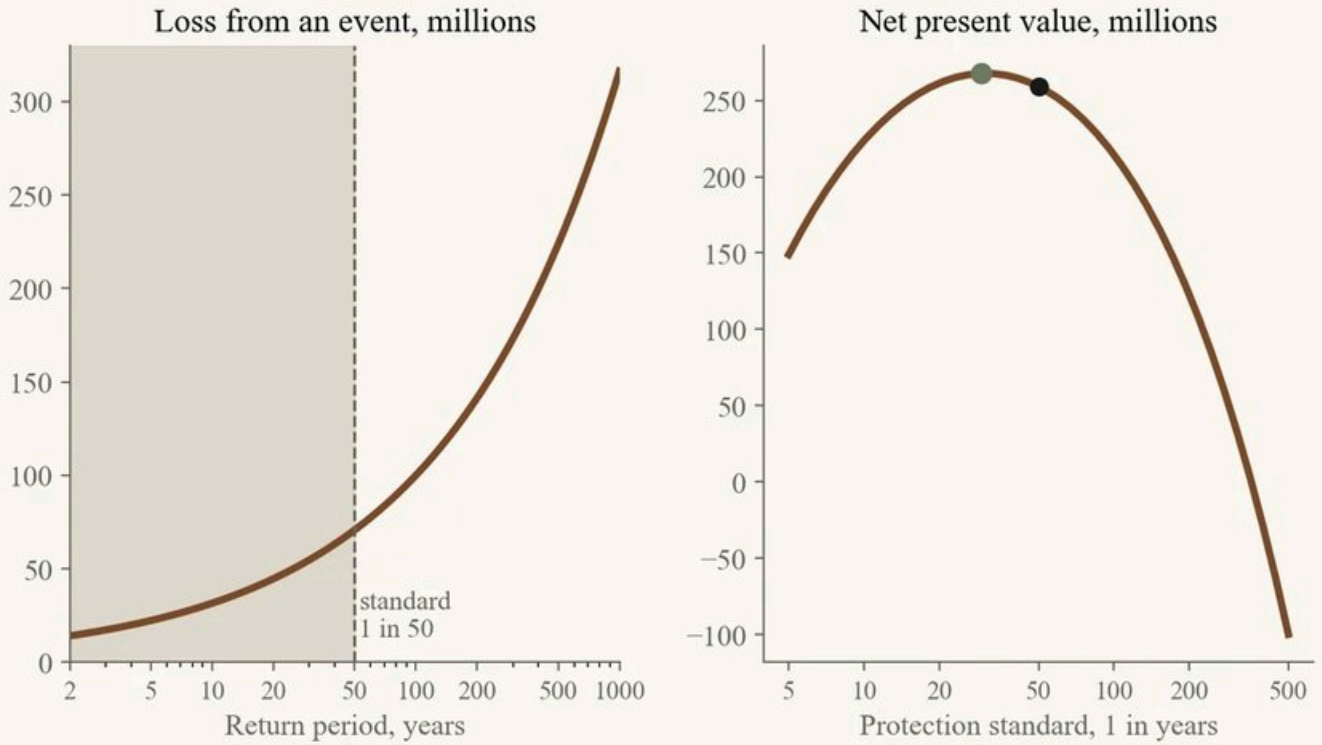
Protection: a standard of R years removes the losses of every event up to R

Drift: frequency and exposed value grow, so avoided loss rises over the years

The appraisal: discount the avoided losses and compare them with cost and upkeep

Author's model (models_w7.py). Risk as hazard, exposure and vulnerability from Kuzma and co-authors (2023), Aqueduct 4.0, pages 10 and 11; benefit-cost range from Hallegatte, Rentschler and Rozenberg (2019), page 12.

Choosing a Protection Standard



The loss: a one-in-100-year event costs 100 million and a one-in-1,000-year event 316 million

The payoff: expected annual loss of 13.5 million, of which a one-in-50 standard avoids 11.3 million

The result: benefit-cost ratio of 2.8 and net present value of 259 million, with the best standard about one in 29

The sensitivity: a heavy tail and fast drift move the best standard to about one in 67

Author's model (models_w7.py); illustrative inputs. Risk elements from Kuzma and co-authors (2023), Aqueduct 4.0; benefit-cost range from Hallegatte, Rentschler and Rozenberg (2019), Overview, page 12.

Three Models Compared

Model	Explains	Prediction	Evidence and limit
Pastor, Stambaugh and Taylor (2021)	What the market pays for sustainability	Greener assets cost more and earn less; the effect needs dispersion in tastes	6 basis point green bond premium; raters disagree; one period
Treasury schedules, June 2026 review, gamma	How a distant benefit is valued	The schedule matters most for late benefits	Only structural uncertainty gives a strongly declining rate; tables stop at 125 years
Expected annual loss with protection	What protection earns	Net benefit peaks at a standard that moves with the tail, drift and schedule	Lifelines range, above 1 in 96 per cent of scenarios; the curve is assumed

Pastor, Stambaugh and Taylor (2021); Baker and co-authors (2018); Berg, Koelbel and Rigobon (2019); HM Treasury (2026); Freeman and Groom (2026); Hallegatte, Rentschler and Rozenberg (2019), page 12.

Case Study: Water Infrastructure and the Environmental Dimension of ESG (Andreas Jobst, Central Bank of the UAE)

The setting and the decision: what the case turns on

The institution: who decides, and under what rules

Water risk and data: what was measured, and how

The financial question: costs, benefits, and who bears them

Discussion: questions for the guest

Summary

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